

REINFORCE Pro Max: Causal Prefix Filtering and Adaptive Advantage Normalization for Critic-Free LLM-RL

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Abstract

Critic-free reinforcement learning for language models must control both token-mass imbalance in sampled advantages and the mismatch between rollout and training policies. We present REINFORCE Pro Max, which combines leave-one-out advantages, sign-preserving asymmetric normalization, and a prefix-cumulative importance-sampling gate. The gate is computed from the cached rollout/snapshot ratio but gradients flow only through the current/snapshot PPO ratio. On a finite autoregressive tree, we prove an exact decomposition of the implemented standard-PPO objective into the raw surrogate, gate and normalization corrections, and a nonnegative clipping deduction. A finite-tree adaptive bound then gives a conditional sufficient certificate for reward improvement. The mathematical statements and their finite-tree proofs are checked in Lean; the formalization does not claim to verify floating-point neural-network training. The current artifact contains structural synthetic checks only and does not report an LLM benchmark.

1 Problem and scope

Critic-free objectives such as RLOO and GRPO are widely used for reasoning tasks [Ahmadian et al., 2024, Shao et al., 2024], while PPO remains a standard clipped policy-gradient baseline [Schulman et al., 2017]. Recent long-horizon trust-region analyses motivate retaining per-position causal structure [Li et al., 2025]. LLM-RL rollouts are generated by π_{roll} while the actor is updated from a frozen snapshot π_{old} . For response $y = (y_1, \dots, y_T)$, write $\rho_t = \pi_\theta(y_t \mid c_t) / \pi_{\text{roll}}(y_t \mid c_t)$ and $w_t = \pi_{\text{old}}(y_t \mid c_t) / \pi_{\text{roll}}(y_t \mid c_t)$. The analyzed configuration is standard PPO (policy_loss_type=ppo), vllm_is_correction_type=reinforce_pro, no dual clip, no KL or entropy term, and sparse terminal reward with zero KL coefficient. These restrictions are essential: optional dual clipping and auxiliary losses change the objective.

We study a finite response alphabet and finite horizon with mutual support, bounded reward $|R| \leq R_{\text{max}}$, and explicitly stated trust-region or moment assumptions. The results are conditional certificates, not guarantees that an arbitrary trained neural policy satisfies those assumptions.

2 Method

For a prompt group of n responses, the raw leave-one-out signal is $A_i = R_i - (n-1)^{-1} \sum_{j \neq i} R_j$. After token expansion, positive and negative active tokens need not have equal mass. Let P, N be the positive and negative token sets. The ideal ProMax factors are chosen so that $\sum_P \alpha A + \sum_N \beta A = 0$ and $|P|^{-1}|N|^{-1}$ -free second-moment normalization equals one on nonzero tokens, with $\alpha, \beta > 0$. Thus signs and within-sign ordering are preserved. The implementation adds an epsilon, cap, clamp, and fallback; the formal result treats their moment effects as explicit residuals.

For each active token, the prefix proxy is

$$M_t^{\text{pre}} = \mathbf{1} \left\{ \ell_- \leq \frac{1}{t} \sum_{s \leq t} \log w_s \leq \ell_+ \right\}, \quad w_t = \exp(\log \pi_{\text{old}} - \log \pi_{\text{roll}}).$$

The implemented loss is

$$\mathcal{L} = \text{MaskedMean}_t \left[M_t^{\text{pre}} \text{stopgrad}(w_t) \left(-\min(r_t \hat{A}_t, \text{clip}(r_t) \hat{A}_t) \right) \right], \quad r_t = \frac{\pi_\theta}{\pi_{\text{old}}}.$$

The prefix gate, cached importance coefficient, action mask, and normalized advantage are detached. Only r_t carries policy gradient.

3 Formal results

Let k index represented response/token positions, let $\omega_k \geq 0$ be the chosen reduction weight, M_k the gate, r_k and w_k the two probability ratios, A_k the raw signal, and H_k the implemented normalized signal. For standard PPO define

$$U = \sum_k \omega_k r_k w_k A_k, \quad G = \sum_k \omega_k r_k w_k (1 - M_k) A_k, \quad N = \sum_k \omega_k r_k w_k M_k (H_k - A_k),$$

and

$$C = \sum_k \omega_k M_k w_k (r_k H_k - \min(r_k H_k, \text{clip}(r_k) H_k)).$$

Then the finite-batch identity is

$$-\mathcal{L} = U - G + N - C, \quad C \geq 0.$$

The gate and normalization terms are signed corrections. The nonnegativity of C is a standard-PPO fact and does not extend to dual clip; a Lean-checked counterexample is included in `formal/ProMaxObjective.lean`.

For a finite autoregressive tree, if every rollout-reachable history of length less than T satisfies either a local TV bound ε or a local reverse-KL bound δ , the future continuation discrepancy after history length t is bounded by

$$\min\{1, (T - t)\varepsilon, \sqrt{(T - t)\delta/2}\}.$$

With bounded rewards, summing these continuation errors bounds the difference between the raw surrogate and the true reward change. Consequently, a corrected empirical ProMax gain exceeding the explicit gate, normalization, clipping, sampling, and adaptive remainder terms is sufficient for positive true reward gain. The statement is for independent evaluation blocks and a fixed finite candidate family, or a pre-specified class with a fixed finite cover.

The finite-tree identities, adaptive recursion, RLOO moments, reduction bounds, and binary-length covariance conditions are checked by the Lean files listed in `formal/PROOF_AUDIT.md`. The audit currently covers 31 labeled mathematical environments and 911 project theorems.

For binary rewards, the ideal normalization has two class amplitudes whose ratio is the ratio of class-conditional mean response lengths. With the actual epsilon stabilizer, cap, clamp, fallback, and token denominator, the complete on-policy update remains a positive multiple of the success-probability gradient when active length depends only on the reward class and both classes have positive mass. If length also varies within a reward class, the update decomposes into this class multiplier plus a within-class covariance error; a strict local-improvement conclusion then requires an explicit covariance margin. The theorem does not claim that arbitrary content-dependent lengths or an arbitrary Adam step improve reward.

4 Implementation correspondence

The code path uses active-token masking and distinguishes the differentiable PPO ratio from the detached rollout correction. Standard PPO is fixed to the configuration above; GSPO, dual clip, KL loss, entropy loss, MoE auxiliary loss, reward shaping, and dynamic group filtering require separate objective terms and are outside the stated certificate. Variable-length dynamic microbatch response-count weighting does not automatically equal a global token mean, so the population reduction must be reported explicitly.

5 Evaluation status and limitations

The current repository includes deterministic synthetic structural checks for normalization and prefix masking. It does not contain matched LLM training, multiple seeds, an independent AIME/MATH-style test set, or a benchmark claim. The formal proofs use exact real arithmetic on finite models; they do not formalize floating-point overflow, GPU kernels, Adam, Ray/vLLM execution, or the assumptions of a general neural policy. These are required before claiming empirical ICLR readiness.

References

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